A mature sporangium bears a multicellular stalk and a spherical or elliptical single layered structure called **capsule**. The capsule contains haploid spores. The wall of the capsule is differentiated into thick walled annulus and thin walled **stomium**. On maturity the sporangium bursts and spores are released. The spores germinate and undergo repeated division to produce a **prothallus**. The prothallus is flat, green and heart shaped. It is monoecious and represents the gametophytic phase. Sex organs called **antheridia** and **archegonia** develop on the prothallus. Antheridia release multiflagellate antherozoids which swim in water and reach the egg of the archegonium to accomplish fertilization. The fertilization results in zygote(2n) and it represents the first cell of sporophytic generation. The zygote develops into embryo which further differentiates into sporophyte. Thus *Adiantum* shows alternation of generation (Figure 2.36).

2.5.6 Types of Stele

The term stele refers to the central cylinder of vascular tissues consisting of xylem, phloem, pericycle and sometimes medullary rays with pith (Figure 2.37).

There are two types of steles

1. Protostele
2. Siphonostele

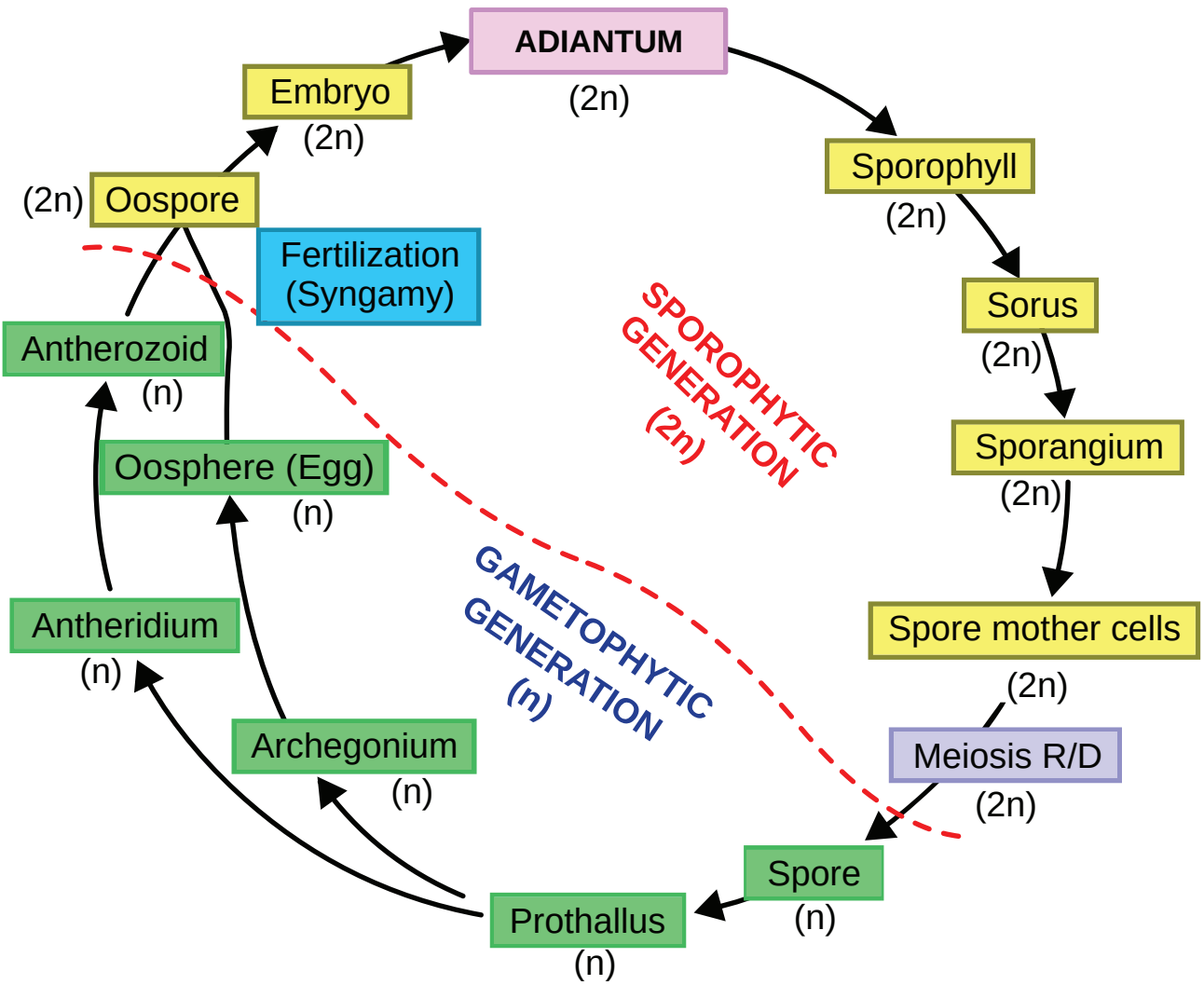


Figure 2.36: Life cycle of *Adiantum*